3ART12 TOUCH

2026-05-01

Variant: CHECKED

Rev 2.1.0+ (Unreleased)

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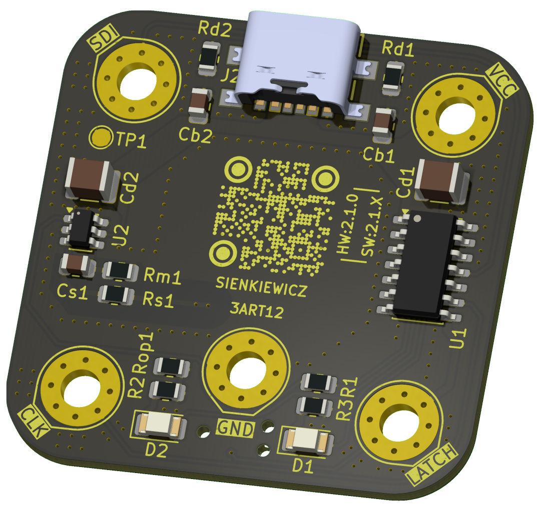
TOP VIEW



BOTTOM VIEW

NOTES

Comment

Not fitted components are marked as ✗

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 01-May-2026

DESIGN CONSIDERATIONS

DESIGN NOTE: Example text for informational design notes.

DESIGN NOTE: Example text for debug notes.

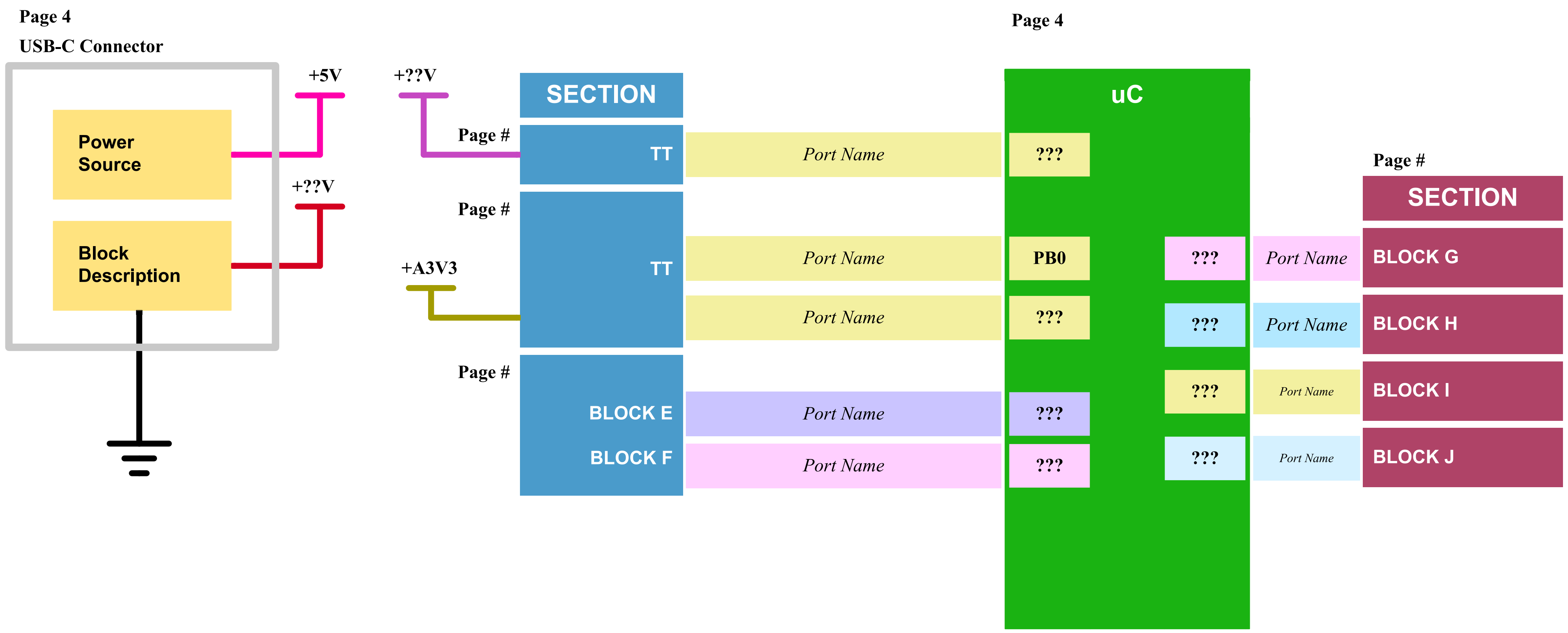
DESIGN NOTE: Example text for cautionary design notes.

DESIGN NOTE: Example text for critical design notes.

LAYOUT NOTE: Example text for critical layout guidelines.

Comments:	Company: BaJRan imgaginary industries		Variant: CHECKED	Git Hash: 6405aa0
	Board Name: 3ART12 TOUCH		Project Name: Sienkiewicz	
Sheet Title:	File Name: Sienkiewicz_TOUCH.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.0+ (Unreleased)
Sheet Path: /		Reviewer:	Size: A3	Sheet: 1 of 9

[2] Block Diagram

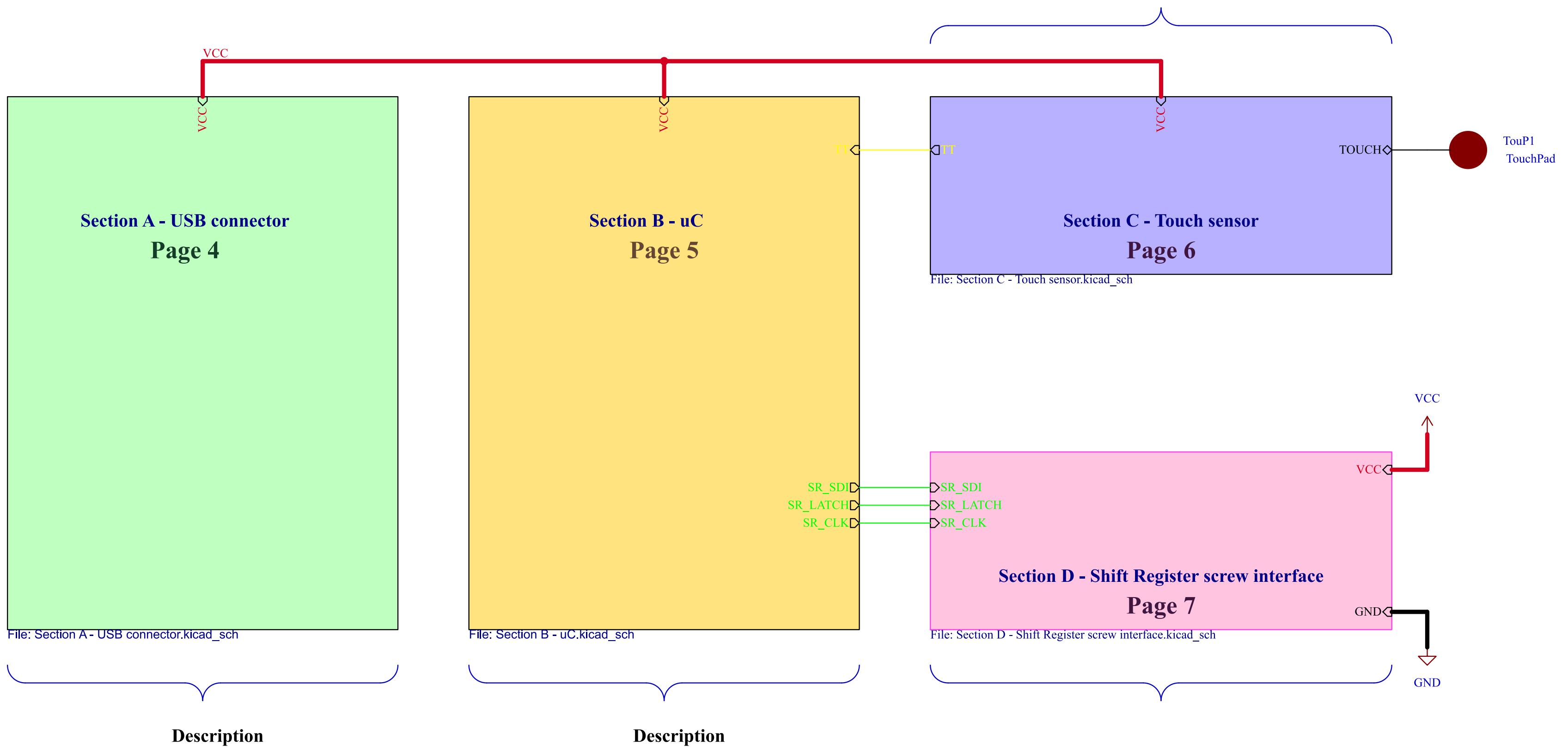


Target specifications:

Input voltage:	?? - ?? V
Spec 2	??
Spec 3	??
Spec 4	??

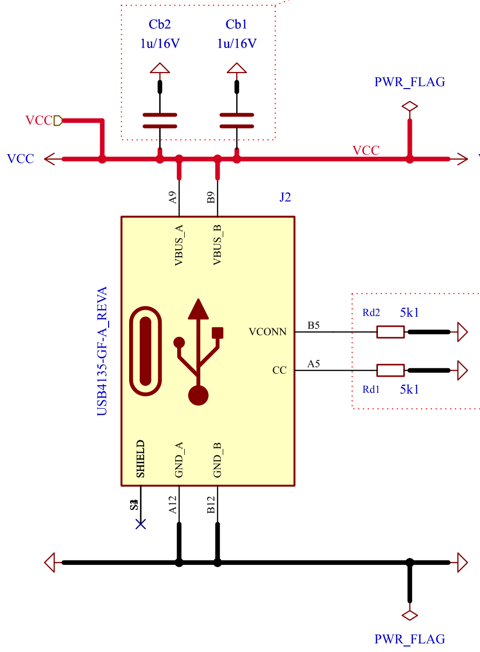
Comments:	Company: BaJRan imgaginary industries		Variant: CHECKED	Git Hash: 6405aa0
	Board Name: 3ART12 TOUCH		Project Name: Sienkiewicz	
Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.0+ (Unreleased)
Sheet Path: /Block Diagram/		Reviewer:	Size: A3	Sheet: 2 of 9

[3] Project Architecture



Comments:	Company: BaJRan imgaginary industries		Variant: CHECKED	Git Hash: 6405aa0
	Board Name: 3ART12 TOUCH		Project Name: Sienkiewicz	
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.0+ (Unreleased)
Sheet Path: /Project Architecture/		Reviewer:	Size: A3	Sheet: 3 of 9

[4] USB connector



DESIGN NOTE:
Cb1, Cb2
bulk capacitor

Release 2.0
August 2019

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USB Type-C Cable and
Connector Specification

Table 4-2 VBUS Source Characteristics

	Minimum	Maximum	Notes
VBUS Leakage Impedance	72.4 kΩ		Leakage between VBUS pins and GND pins on receptacle when VBUS is not being sourced.
VBUS Capacitance		3000 µF	Capacitance for source-only ports between VBUS and GND pins on receptacle when VBUS is not being sourced.
		10 µF	Capacitance for DRP ports between VBUS and GND pins on receptacle when VBUS is not being sourced.

DESIGN NOTE:
Per USB-IF specification, a sink role is established by placing 5.1 kΩ pull-down (Rd) resistors on CC1 and CC2.

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USB Type-C Cable and
Connector Specification

Table 4-25 Sink CC Termination (Rd) Requirements

Rd Implementation	Nominal value	Can detect power capability?	Max voltage on pin
± 20% voltage clamp[1]	1.1 V	No	1.32 V
± 20% resistor to GND	5.1 kΩ	No	2.18 V
± 10% resistor to GND	5.1 kΩ	Yes	2.04 V

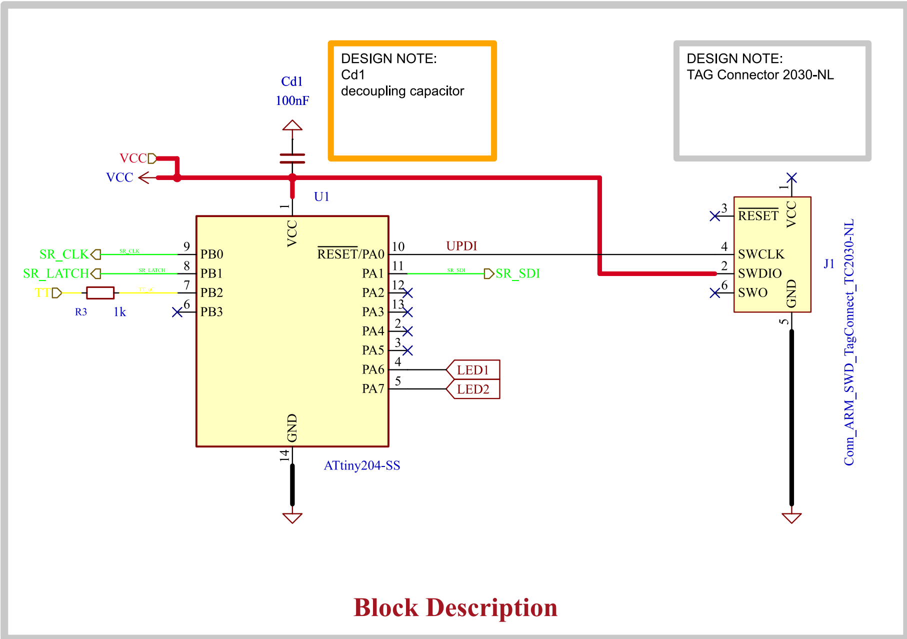
Note:
1. The clamp implementation inhibits USB PD communication although the system can start with the clamp and transition to the resistor once it is able to do USB PD.

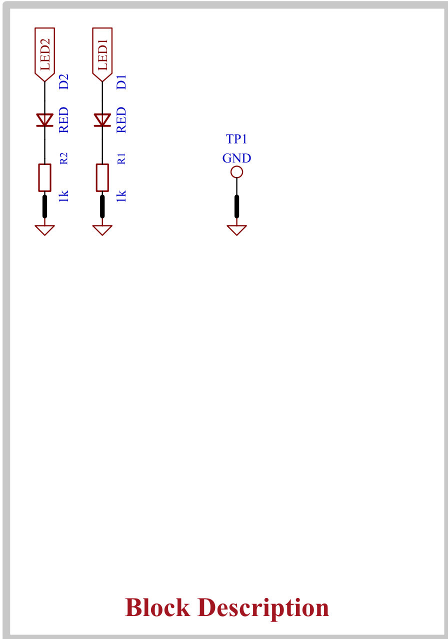
www.usb.org/sites/default/files/USB%20Type-C%20Spec%20R2.0%20-%20August%202019.pdf

LAYOUT NOTE:
blablabla

Comments: AN5346 STM32G474 Datasheet p.81 J. Pieper ADC investigation	Company: BaJRan imgaginary industries		Variant: CHECKED	Git Hash: 6405aa0
	Board Name: 3ART12 TOUCH		Project Name: Sienkiewicz	
Sheet Title: USB connector	File Name: Section A - USB connector.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.0+ (Unreleased)
Sheet Path: /Project Architecture/Section A - USB connector/		Reviewer:	Size: A4	Sheet: 4 of 9

[5] uC & prog connector





Description

Ideal for both debugging and production programming of AVR MCUs using the Microchip PICkit 4/5 or SNAP in-circuit debuggers, the 6 pin plug-of-nails™ cable terminates in a 8 pin 0.1" male SIP strip for connection to the debugger. The target board pin connections on the plug-of-nails™ connector are the same as on our TC2030-ICESPI cable. The cable supports programming via PDI, UPDI, aWire, debugWire, SPI or TDI.

Note: For AVR processors – DOES NOT WORK WITH PIC MCU

TC2030-PKT-ICESPI Connections

8 Pin SIP	SPI	TPI	UPDI	PDI	aWire	debugWIRE	6-pin TC2030
1							
2	VTG	VTG	VTG	VTG	VTG	VTG	2
3	GND	GND	GND	GND	GND	GND	6
4	MISO	DAT	PDI_DATA	PDI_DATA	DATA		1
5	SCK	CLK					3
6		RST		PDI_CLK		RESET	5
7	MOSI						4
8							

References:

Flexible I/O worked examples
Flexible I/O source configuration

Comments:	Company:		Variant:	Git Hash:
	BaJRan imgaginary industries		CHECKED	6405aa0
	Board Name: **3ART12 TOUCH**		Project Name: **Sienkiewicz**	
Sheet Title: uC & prog connector	File Name: Section B - uC.kicad_sch	Designer: Pawel Baran	Date: 2026-04-16	Revision: 2.1.0+ (Unreleased)
Sheet Path: /Project Architecture/Section B - uC/		Reviewer:	Size: **A4**	Sheet: **5** of **9**

[6] Touch Sensor

DESIGN NOTE:

Table 3-2. Predefined Auto-off Delay (Active High Output)

Vt	Auto-off Delay (t_o)
Vss	Infinity (remain on until toggled to off)
Vdd	15 minutes
OUT	60 minutes

Table 3-3. Predefined Auto-off Delay (Active Low Output)

Vt	Auto-off Delay (t_o)
Vss	15 minutes
Vdd	Infinity (remain on until toggled to off)
OUT	60 minutes

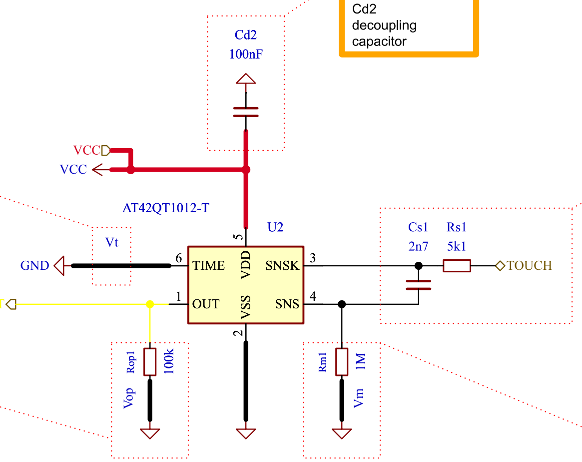
DESIGN NOTE:

Output Polarity Selection

The output (OUT pin) of the QT1012 can be configured to have an active high or active low output by means of the output configuration resistor Rop. The resistor is connected between the output and either Vss or Vdd (see Figure 3-4 and Table 3-1). A typical value for Rop is 100 kΩ.

Table 3-1. Output Configuration

Name (Vop)	Function (Output Polarity)
Vss	Active high
Vdd	Active low



DESIGN NOTE:
Cd2
decoupling
capacitor

ACTIVE HIGH OUTPUT

AUTO DELAY OFF

DESIGN NOTE:
sense circuit
Cs [2nF - 60nF]
Rs [4k7 - 20k]

DESIGN NOTE:

A single 1 MΩ resistor (Rm) is connected between the SNS pin and the logic level Vm to provide three auto-off functions: delay multiplication, delay override and delay retriggering. On power-up the logic level at Vm is assessed and the delay multiplication factor is set to x1 or x24 accordingly (see Figure 3-6 on page 12, Table 3-2 on page 12 and Table 3-3 on page 12). At the end of each acquisition cycle the logic level of Vm is monitored to see if an Auto-off delay override is required (see Section 3.11.5 on page 17).

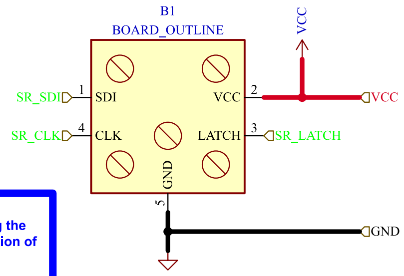
Setting the delay multiplier to x24 will decrease the key sensitivity. To compensate, it may be necessary to increase the value of Cs.

Figure 3-6. Predefined Delay
Table 3-4. Auto-off Delay Multiplier

Vm	Auto-off Delay Multiplier
Vss	$t_o \times 1$
Vdd	$t_o \times 24$

Comments:	Company: BaJRan imgaginary industries		Variant: CHECKED	Git Hash: 6405aa0
	Board Name: 3ART12 TOUCH		Project Name: Sienkiewicz	
Sheet Title: Touch Sensor	File Name: Section C - Touch sensor.kicad_sch	Designer: Pawel Baran	Date: 2026-04-14	Revision: 2.1.0+ (Unreleased)
Sheet Path: /Project Architecture/Section C - Touch sensor/		Reviewer:	Size: A4	Sheet: 6 of 9

[7] Shift Register screw interface

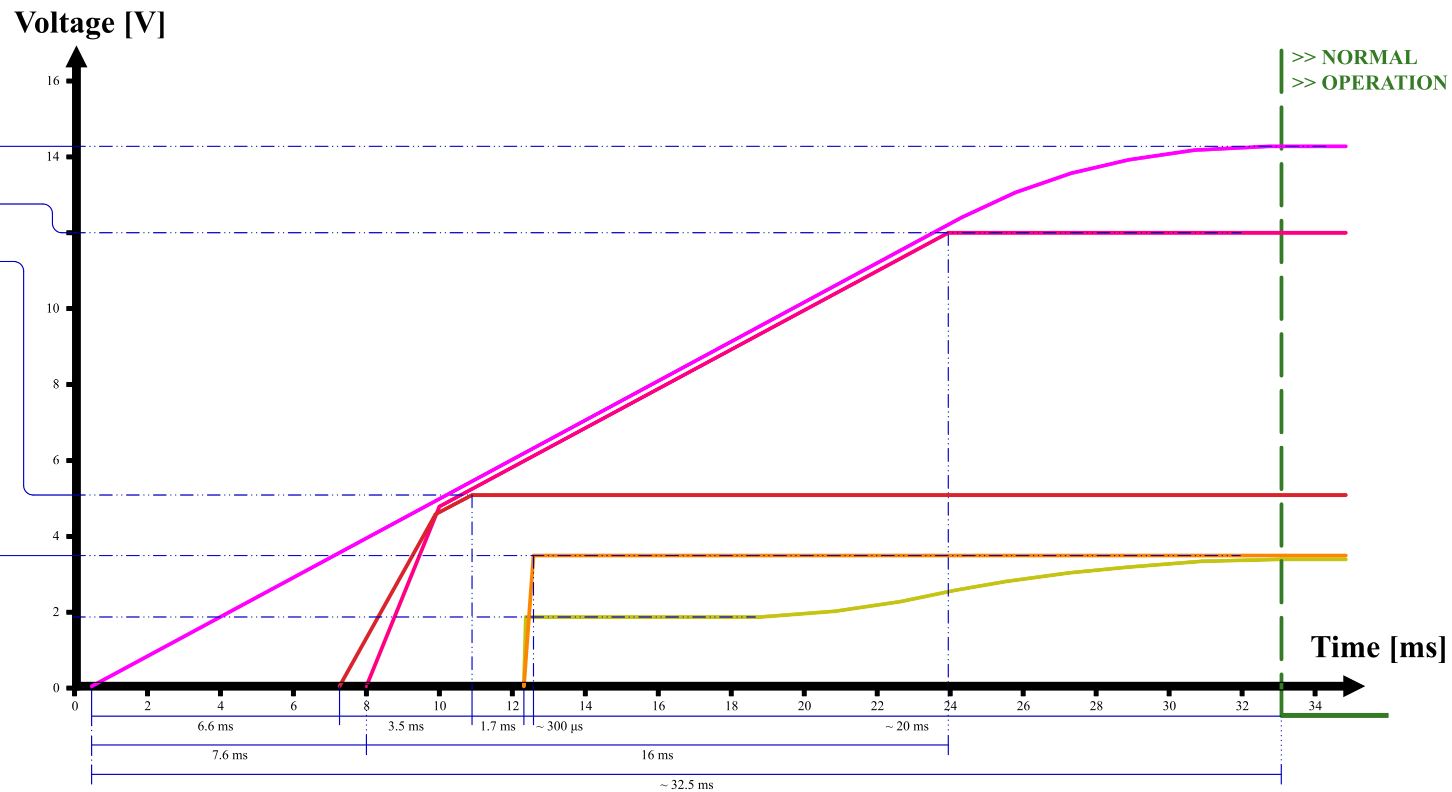


LAYOUT NOTE:
Template illustrating the mechanical connection of two PCBs using M3 screws.

Comments:	Company:		Variant: CHECKED	Git Hash: 6405aa0
	Board Name: 3ART12 TOUCH		Project Name: Sienkiewicz	
Sheet Title: Shift Register screw interface	File Name: Section D - Shift Register screw interface.kicad_sch	Designer: Pawel Baran	Date: 2026-04-16	Revision:
Sheet Path: /Project Architecture/Section D - Shift Register screw interface/		Reviewer:	Size: A4	Sheet: 7 of 9

[8] Power - Sequencing

NAME	SOURCE	LEVEL
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%



Comments:	Company: BaJRan imgaginary industries	Variant: CHECKED	Git Hash: 6405aa0
	Board Name: **3ART12 TOUCH**	Project Name: **Sienkiewicz**	
Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12
Revision: 2.1.0+ (Unreleased)			
Sheet Path: /Power - Sequencing/	Reviewer:	Size: **A4**	Sheet: **8** of **9**

[9] Revision History

Version 2.0.0 - 2026-04-24

Added

- First 2.0.0 Version of pcb and schematic

Version 2.0.1 - 2026-04-25

Changed

- Modified silkscreen look, by adding bigger QR-code. small rearangment.

Version 2.0.2 - 2026-04-26

Changed

- Modified silkscreen look
- Fabrication layers update

Version 2.1.0 - 2026-05-01

Changed

- B-B connector maping changes, CLK and SDI swapped (fixes #4)

Comments:	Company: BaJRan imgaginary industries		Variant: CHECKED	Git Hash: 6405aa0
	Board Name: 3ART12 TOUCH		Project Name: Sienkiewicz	
Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.0+ (Unreleased)
Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 9 of 9